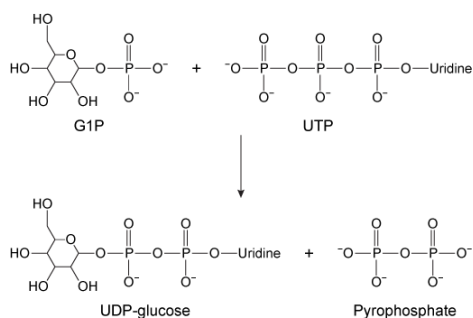


The reaction between UTP and glucose-1-phosphate (G1P) to form UDP-glucose in glycogenesis is shown below:



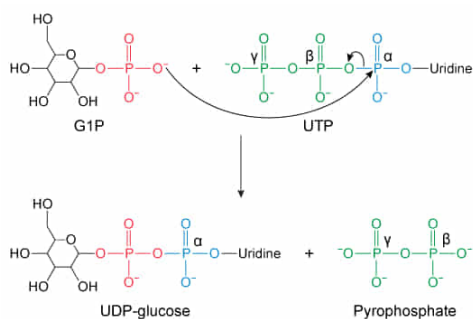
Which phosphates from the reactants are present in UDP-glucose?

- ☐ A. The α- and β-phosphates from UTP [14%]
- ☐ B. The β- and γ-phosphates from UTP [6%]
- ☐ C. The γ-phosphate from UTP and the C1 phosphate from G1P [22%]
- ☒ D. The α-phosphate from UTP and the C1 phosphate from G1P [56%]

Correct

56% answered correctly

Explanation:



Anabolism is the synthesis of complex biological molecules from simpler constituents, **and it requires energy input**. The energy needed for anabolic processes is generally provided by cleaving **high-energy phosphodiester bonds in nucleotide triphosphates (NTPs)** such as ATP, GTP, or UTP.

The three phosphate groups on an NTP are designated as the **α-, β-, and γ-phosphates**, with the α-phosphate being linked to the nucleoside and the γ-phosphate being farthest away from the nucleoside. NTP cleavage typically releases either the γ-phosphate as inorganic phosphate (P_i) or the γ- and β-phosphates as pyrophosphate.

Glycogenesis is the anabolic synthesis of glycogen from glucose molecules. In the second-to-last step of glycogenesis, the C1 phosphate of G1P acts as a nucleophile to attack the α-phosphate of UTP. The bond

between the α- and β-phosphates breaks, and the **β- and γ-phosphates are released as pyrophosphate**. Accordingly, the two phosphates that are incorporated into UDP-glucose are the **α-phosphate from UTP** and the **C1 phosphate of G1P**.

(Choices A and B) Following the reaction shown, the α-phosphate from UTP is present in UDP-glucose, but the β- and γ-phosphates are released as pyrophosphate.

(Choice C) The γ-phosphate is the furthest phosphate from uridine in UTP. It is part of the pyrophosphate leaving group, not UDP-glucose.

Educational objective:

Anabolic processes require energy input, which is typically provided by cleaving phosphodiester bonds in nucleotide triphosphates (NTPs) such as ATP, GTP, or UTP. These cleavage events either release the γ -phosphate as inorganic phosphate (P_i) or they release the γ - and β -phosphates as pyrophosphate. The α -phosphate remains attached to the NTP.

Time spent:0 sec**Subject:**
Biochemistry**QID:**402439**System:**
Metabolic Reactions